

MA1521 Calculus for Computing

Lecture Group 2

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Lesson 05: Tuesday 25 August 2026

(Week 3)

Remark. Q04.12 to Q04.15 are leftover exercises from the Friday class in Week 2.

Remark. Q05.1 to Q05.5 are on Section 11.5 of Thomas' Calculus.

Q05.6 to Q05.9 are on Section 10.3 -10.4.

Q05.10 to Q05.14 are on Section 12.1 -12.3.

Alert: Please note the two recent announcements in Canvas:

- List of Approved Calculators for test and exam
- Midterm Test

From our solution of **Q04.11**, we saw that to write down an equation of a plane $\mathcal{P}$, we typically need the data of a point $(\mathbf{x}_o, \mathbf{y}_o, \mathbf{z}_o)$ on $\mathcal{P}$, and a normal vector $\vec{\mathbf{N}} = \langle \mathbf{a}, \mathbf{b}, \mathbf{c} \rangle$ to $\mathcal{P}$. Then an equation of $\mathcal{P}$ is given by

$$\begin{aligned} & \langle \mathbf{x} - \mathbf{x}_o, \mathbf{y} - \mathbf{y}_o, \mathbf{z} - \mathbf{z}_o \rangle \cdot \langle \mathbf{a}, \mathbf{b}, \mathbf{c} \rangle = 0 \\ \iff & \mathbf{a}(\mathbf{x} - \mathbf{x}_o) + \mathbf{b}(\mathbf{y} - \mathbf{y}_o) + \mathbf{c}(\mathbf{z} - \mathbf{z}_o) = 0 \\ \iff & \mathbf{a}\mathbf{x} + \mathbf{b}\mathbf{y} + \mathbf{c}\mathbf{z} - \mathbf{d} = 0 \end{aligned}$$

where $\mathbf{d} = \mathbf{a}\mathbf{x}_o + \mathbf{b}\mathbf{y}_o + \mathbf{c}\mathbf{z}_o$.

Remark. Two linear equations represent the same plane if and only if their coefficients are proportional. For example, the equations

$$\mathbf{x} + 2\mathbf{y} + 3\mathbf{z} + 5 = 0 \quad \text{and} \quad 2\mathbf{x} + 4\mathbf{y} + 6\mathbf{z} + 10 = 0$$

represent the same plane

Q04.12. Consider the plane with equation $x + 3y - 7z - 8 = 0$. Find a normal vector of the plane.

Solution:

$$\vec{N} = 1 \cdot i + 3j - 7k$$

Q04.13. Consider the two following two planes:

$$\mathbf{P}_1 : x + 3y - 7z - 8 = 0,$$

$$\mathbf{P}_2 : -2x - 6y + 14z = 5.$$

Are the two planes $\mathbf{P}_1$ and $\mathbf{P}_2$ parallel? Justify your answer.

Solution: Consider the planes

$$P_1 : x + 3y - 7z - 8 = 0,$$

$$P_2 : -2x - 6y + 14z = 5.$$

A normal vector of P_1 is $\vec{N}_1 = \langle 1, 3, -7 \rangle$.

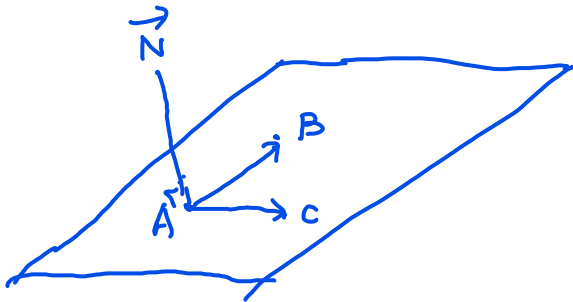
A normal vector of P_2 is $\vec{N}_2 = \langle -2, -6, -14 \rangle$.

Since $\vec{N}_2 = -2\vec{N}_1$, it follows that $\vec{N}_2$ and $\vec{N}_1$ are parallel.

Thus the two planes P_1 and P_2 are parallel.

Q04.14. Find an equation of the plane passing through the points $A(3, 3, 0)$, $B(3, 0, 1)$ and $C(0, 2, 1)$.

Suggestion: First find two vectors parallel to the plane, and then find a vector normal to the plane.



Solution: Since $\mathbf{A}(3, 3, 0)$, $\mathbf{B}(3, 0, 1)$ and $\mathbf{C}(0, 2, 1)$ lie on the plane $\mathbf{P}$, the vectors

$$\overrightarrow{\mathbf{AB}} = \langle 3 - 3, 0 - 3, 1 - 0 \rangle = \langle 0, -3, 1 \rangle,$$

$$\overrightarrow{\mathbf{AC}} = \langle 0 - 3, 2 - 3, 1 - 0 \rangle = \langle -3, -1, 1 \rangle$$

are parallel to $\mathbf{P}$. Thus a normal vector of $\mathbf{P}$ is given by

$$\overrightarrow{\mathbf{N}} = \overrightarrow{\mathbf{AB}} \times \overrightarrow{\mathbf{AC}}$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -3 & 1 \\ -3 & -1 & 1 \end{vmatrix}$$

$$= \mathbf{i} \begin{vmatrix} -3 & 1 \\ -1 & 1 \end{vmatrix} - \mathbf{j} \begin{vmatrix} 0 & 1 \\ -3 & 1 \end{vmatrix} + \mathbf{k} \begin{vmatrix} 0 & -3 \\ -3 & -1 \end{vmatrix}$$

$$= -2\mathbf{i} - 3\mathbf{j} - 9\mathbf{k}.$$

Using the normal vector $\vec{N} = -2\mathbf{i} - 3\mathbf{j} - 9\mathbf{k}$ and the point $\mathbf{A}(3, 3, 0)$ of P , one gets an equation of P given by

$$\langle \mathbf{x} - 3, \mathbf{y} - 3, \mathbf{z} - 0 \rangle \cdot \vec{N} = 0$$

$$\iff \langle \mathbf{x} - 3, \mathbf{y} - 3, \mathbf{z} - 0 \rangle \cdot (-2\mathbf{i} - 3\mathbf{j} - 9\mathbf{k}) = 0$$

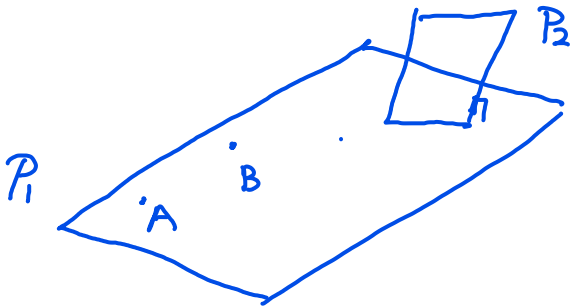
$$\iff -2 \cdot (\mathbf{x} - 3) - 3(\mathbf{y} - 3) - 9 \cdot \mathbf{z} = 0$$

$$\iff -2\mathbf{x} - 3\mathbf{y} - 9\mathbf{z} + 15 = 0$$

$$\iff 2\mathbf{x} + 3\mathbf{y} + 9\mathbf{z} = 15.$$

Q04.15. Find an equation of the plane P_1 that passes through the points $A = (-1, 1, -1)$ and $B = (2, 2, 1)$, and is perpendicular to the plane $P_2 : 4x - 6y + 2z = 1$.

Suggestion: First find two vectors parallel to the plane P_1 .



Solution: Since $\mathbf{P}_1$ that passes through the points $\mathbf{A} = (-1, 1, -1)$ and $\mathbf{B} = (2, 2, 1)$, it follows that the vector

$$\overrightarrow{\mathbf{AB}} = \langle 2 - (-1), 2 - 1, 1 - (-1) \rangle = \langle 3, 1, 2 \rangle$$

is parallel to $\mathbf{P}$. Note that a normal vector of

$\mathbf{P}_2 : 4x - 6y + 2z = 1$ is $\overrightarrow{\mathbf{N}}_2 = \langle 4, -6, 2 \rangle$.

Since $\mathbf{P}_2$ is perpendicular to $\mathbf{P}_1$, it follows that $\overrightarrow{\mathbf{N}}_2$ is parallel to $\mathbf{P}_1$. Hence a normal vector of $\mathbf{P}_1$ is given by

$$\overrightarrow{\mathbf{N}}_1 = \overrightarrow{\mathbf{AB}} \times \overrightarrow{\mathbf{N}}_2$$

$$\begin{aligned} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 1 & 2 \\ 4 & -6 & 2 \end{vmatrix} = \mathbf{i} \begin{vmatrix} 1 & 2 \\ -6 & 2 \end{vmatrix} - \mathbf{j} \begin{vmatrix} 3 & 2 \\ 4 & 2 \end{vmatrix} + \mathbf{k} \begin{vmatrix} 3 & 1 \\ 4 & -6 \end{vmatrix} \\ &= 14\mathbf{i} + 2\mathbf{j} - 22\mathbf{k}. \end{aligned}$$

Using the normal vector $\vec{N}_1 = 14\mathbf{i} + 2\mathbf{j} - 22\mathbf{k}$ and the point $\mathbf{A} = (-1, 1, -1)$ of P_1 , one gets an equation of P_1 given by

$$\langle \mathbf{x} - (-1), \mathbf{y} - 1, \mathbf{z} - (-1) \rangle \cdot \vec{N}_1 = 0$$

$$\iff \langle \mathbf{x} + 1, \mathbf{y} - 1, \mathbf{z} + 1 \rangle \cdot (14\mathbf{i} + 2\mathbf{j} - 22\mathbf{k}) = 0$$

$$\iff 14 \cdot (\mathbf{x} + 1) + 2(\mathbf{y} - 1) - 22 \cdot (\mathbf{z} + 1) = 0$$

$$\iff 14\mathbf{x} + 2\mathbf{y} - 22\mathbf{z} - 10 = 0$$

$$\iff 7\mathbf{x} + \mathbf{y} - 11\mathbf{z} - 5 = 0.$$

Lines in $\mathbb{R}^3$. A line L in $\mathbb{R}^3$ is determined by a point $(\mathbf{x}_o, \mathbf{y}_o, \mathbf{z}_o)$ on L and a vector $\vec{\mathbf{v}} = \langle \mathbf{a}, \mathbf{b}, \mathbf{c} \rangle$ parallel to L .

For a general point $(\mathbf{x}, \mathbf{y}, \mathbf{z})$ on L , the vector $\langle \mathbf{x} - \mathbf{x}_o, \mathbf{y} - \mathbf{y}_o, \mathbf{z} - \mathbf{z}_o \rangle$ is parallel to $\vec{\mathbf{v}}$, and thus there exists some real number t such that

$$\begin{aligned} \langle \mathbf{x} - \mathbf{x}_o, \mathbf{y} - \mathbf{y}_o, \mathbf{z} - \mathbf{z}_o \rangle &= t \vec{\mathbf{v}} = t \langle \mathbf{a}, \mathbf{b}, \mathbf{c} \rangle \\ \implies \mathbf{x} &= \mathbf{x}_o + \mathbf{a}t, \quad \mathbf{y} = \mathbf{y}_o + \mathbf{b}t, \quad \mathbf{z} = \mathbf{z}_o + \mathbf{c}t. \quad (*) \end{aligned}$$

(parametric equations of L).

Some other ways to write down parametric equations of L :

$$\begin{aligned}\overrightarrow{r(t)} &= \overrightarrow{r_o} + t\overrightarrow{v}, & \overrightarrow{r_o} &= x_o\mathbf{i} + y_o\mathbf{j} + z_o\mathbf{k} \\ &= (x_o + at)\mathbf{i} + (y_o + bt)\mathbf{j} + (z_o + ct)\mathbf{k},\end{aligned}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x_o \\ y_o \\ z_o \end{pmatrix} + t \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} x_o + at \\ y_o + bt \\ z_o + ct \end{pmatrix}$$

Q05.1. Write down parametric equation for the line in $\mathbb{R}^3$ passing through the point $(3, -1, 4)$ and parallel to the vector $2\mathbf{i} + 5\mathbf{j} - 7\mathbf{k}$. Write down also a vector equation for the line.

Solution:

$$\langle x-3, y-(-1), z-4 \rangle = t \langle 2, 5, -7 \rangle$$

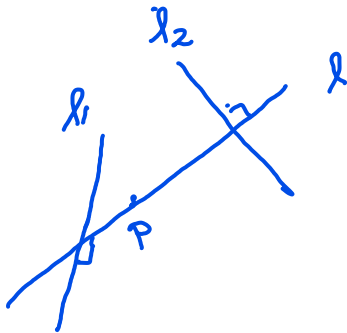
$$x = 3 + 2t, \quad y = -1 + 5t, \quad z = 4 - 7t.$$

$$\vec{r}(t) = (3+2t)\mathbf{i} + (-1+5t)\mathbf{j} + (4-7t)\mathbf{k}.$$

Q05.2. Find parametric equations for the line in $\mathbb{R}^3$ through the point $P(4, 2, 1)$ and perpendicular to two lines ℓ_1 and ℓ_2 given by:

$$\ell_1 : x = 3t + 2, y = -4t - 1, z = 4t - 9.$$

$$\ell_2 : x = 3t - 4, y = -t + 6, z = 5t.$$



Solution: It is given that the line ℓ passes through the point $\mathbf{P}(4, 2, 1)$ and is perpendicular to two lines ℓ_1 and ℓ_2

It follows from the equation

$$\ell_1 : \mathbf{x} = 3t + 2, \mathbf{y} = -4t - 1, \mathbf{z} = 4t - 9$$

that a direction vector of ℓ_1 is $\mathbf{v}_1 = \langle 3, -4, 4 \rangle$.

Similarly, it follows from the equation

$$\ell_2 : \mathbf{x} = 3t - 4, \mathbf{y} = -t + 6, \mathbf{z} = 5t$$

that a direction vector of ℓ_2 is $\mathbf{v}_2 = \langle 3, -1, 5 \rangle$.

Since ℓ is perpendicular to the lines ℓ_1 and ℓ_2 , it follows that ℓ is perpendicular to $\mathbf{v}_1$ and $\mathbf{v}_2$. Hence the cross product of $\mathbf{v}_1$ and $\mathbf{v}_2$ gives a direction vector of the line ℓ .

So, a direction vector $\mathbf{v}$ of ℓ is

$$\mathbf{v} = \mathbf{v}_1 \times \mathbf{v}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -4 & 4 \\ 3 & -1 & 5 \end{vmatrix} = \langle -16, -3, 9 \rangle.$$

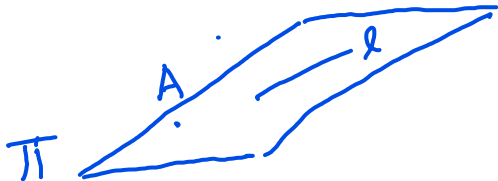
Together with the given fact that ℓ passes through the point $\mathbf{P}(4, 2, 1)$, it follows that parametric equations for ℓ are given by:

$$x = 4 - 16t, \quad y = 2 - 3t, \quad z = 1 + 9t.$$

Q05.3. Let $\mathbf{A} = (2, -1, 3)$, $\mathbf{B} = (4, 1, 2)$ and ℓ be the line with parametric equations

$$x = 3t + 1, \quad y = t + 1, \quad z = -t + 2.$$

- (a) Find an equation of the plane containing $\mathbf{A}$ and ℓ .
- (b) Show that there is no plane that contains $\mathbf{A}$, $\mathbf{B}$ and the line $\ell : x = 3t + 1, y = t + 1, z = -t + 2$.



Solution: (a) Let Π be the required plane that contains $\mathbf{A} = (2, -1, 3)$ and the line

$$\ell : \mathbf{x} = 3t + 1, \mathbf{y} = t + 1, \mathbf{z} = -t + 2. \quad (1)$$

By taking $t = 0$ in (1), we see that the point $\mathbf{C} = (1, 1, 2)$ lies on ℓ , and thus Π contains $\mathbf{C} = (1, 1, 2)$.

Together with the given fact that Π contains $\mathbf{A} = (2, -1, 3)$, it follows that

$$\overrightarrow{\mathbf{CA}} = (2 - 1)\mathbf{i} + (-1 - 1)\mathbf{j} + (3 - 2)\mathbf{k} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$$

is a vector in Π . From (1), one sees that a direction vector of ℓ is given by $\mathbf{v} = 3\mathbf{i} + \mathbf{j} - \mathbf{k}$.

Note that $\mathbf{v}$ is a vector parallel to Π , since Π contains ℓ .

Since $\overrightarrow{CA}$ and $\mathbf{v}$ are vectors parallel to Π , a normal vector of Π can be given by

$$\mathbf{n} = \overrightarrow{CA} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -2 & 1 \\ 3 & 1 & -1 \end{vmatrix} = \mathbf{i} + 4\mathbf{j} + 7\mathbf{k}..$$

It follows that an equation of the plane Π is given by $x + 4y + 7z = d$, where d is a constant.

Since Π contains the point $A = (2, -1, 3)$, it follows that

$$2 + 4 \cdot (-1) + 7 \cdot 3 = d \implies d = 19.$$

Hence an equation of the plane Π is

$$x + 4y + 7z = 19.$$

(b) From part (a), we know that an equation of the plane Π that contains $\mathbf{A}$ and the line ℓ is given by

$$x + 4y + 7z = 19. \quad (1)$$

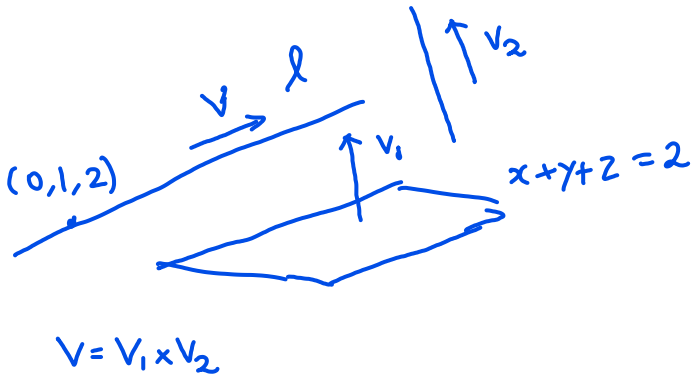
Substituting the coordinates of $\mathbf{B}(4, 1, 2)$ into the left side of (1), we get

$$4 + 4 \cdot 1 + 7 \cdot 2 = 22 \neq 19.$$

Thus $\mathbf{B}$ does not lie on the plane Π .

Hence there does not exist a plane that contains the points $\mathbf{A}$, $\mathbf{B}$ and the line ℓ .

Q05.4. Find parametric equations of the line through the point $(0, 1, 2)$ that is parallel to the plane $x + y + z = 2$ and perpendicular to the line $x = 1 + t$, $y = 1 - t$, $z = 2t$. Write down also a vector equation for the line.



Solution: Since the required line ℓ is parallel to the plane $x + y + z = 2$, the normal vector $\mathbf{v}_1 = \langle 1, 1, 1 \rangle$ of the plane is perpendicular to ℓ .

It is also given that the line $x = 1 + t$, $y = 1 - t$, $z = 2t$ (with direction vector $\mathbf{v}_2 = \langle 1, -1, 2 \rangle$) $\perp \ell$.

Thus a direction vector $\mathbf{v}$ of ℓ is given by

$$\mathbf{v} = \mathbf{v}_1 \times \mathbf{v}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 1 & -1 & 2 \end{vmatrix} = 3\mathbf{i} - \mathbf{j} - 2\mathbf{k}.$$

Together with fact that ℓ contains the point $(0, 1, 2)$, it follows that parametric equations of ℓ are given by

$$x = 3t, \quad y = 1 - t, \quad z = 2 - 2t.$$

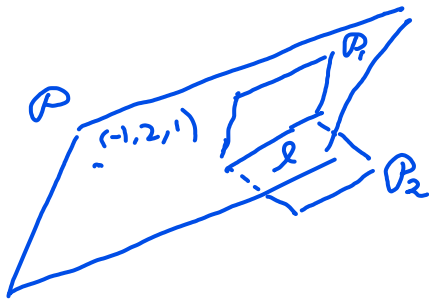
Rewriting the above parametric equations of ℓ given by

$$\mathbf{x} = 3t, \quad \mathbf{y} = 1 - t, \quad \mathbf{z} = 2 - 2t,$$

we get a vector equation of ℓ :

$$\overrightarrow{r(t)} = 3t\mathbf{i} + (1 - t)\mathbf{j} + (2 - 2t)\mathbf{k}.$$

Q05.5. Find an equation of the plane that contains the point $(-1, 2, 1)$ and the line of intersection of the planes $x + y - z = 2$ and $2x - y + 3z = 1$.



Solution: Let $\mathcal{P}$ be the plane containing the point $\mathbf{A}(-1, 2, 1)$ and the line of intersection ℓ of the planes $\mathcal{P}_1 : \mathbf{x} + \mathbf{y} - \mathbf{z} = 2$ and $\mathcal{P}_2 : 2\mathbf{x} - \mathbf{y} + 3\mathbf{z} = 1$.

Note that a point $(\mathbf{x}, \mathbf{y}, \mathbf{z})$ lies on ℓ iff the point satisfies the equations of both $\mathcal{P}_1$ and $\mathcal{P}_2$.

Thus to find ℓ , we solve the equations of $\mathcal{P}_1$ and $\mathcal{P}_2$ simultaneously (and treating one of the variables $\mathbf{z}$ as parameter for the line):

$$\begin{cases} \mathbf{x} + \mathbf{y} = \mathbf{2} + \mathbf{z}, & (1) \\ \mathbf{2x} - \mathbf{y} = \mathbf{1} - \mathbf{3z}. & (2) \end{cases}$$

By adding (1) to (2), we get

$$\mathbf{3x} = \mathbf{3} - \mathbf{2z} \implies \mathbf{x} = \mathbf{1} - \frac{\mathbf{2}}{\mathbf{3}}\mathbf{z}.$$

Substituting this into (1), we get

$$\mathbf{y} = \mathbf{2} + \mathbf{z} - (\mathbf{1} - \frac{\mathbf{2}}{\mathbf{3}}\mathbf{z}) = \mathbf{1} + \frac{\mathbf{5}}{\mathbf{3}}\mathbf{z}.$$

Now letting $\mathbf{z} = \mathbf{t}$, we get parametric equations of ℓ given by

$$\mathbf{x} = \mathbf{1} - \frac{\mathbf{2}}{\mathbf{3}}\mathbf{t}, \quad \mathbf{y} = \mathbf{1} + \frac{\mathbf{5}}{\mathbf{3}}\mathbf{t}, \quad \mathbf{z} = \mathbf{t}.$$

By putting $\mathbf{t} = \mathbf{0}$, we get a point $\mathbf{B(1, 1, 0)}$ on ℓ . Also, a direction vector of ℓ is $-\frac{\mathbf{2}}{\mathbf{3}}\mathbf{i} + \frac{\mathbf{5}}{\mathbf{3}}\mathbf{j} + \mathbf{k}$.

To avoid fractions (and for illustration), we may also take another direction vector of ℓ given by $\mathbf{v} = 2\mathbf{i} - 5\mathbf{j} - 3\mathbf{k}$.

Since the plane $\mathcal{P}$ contains ℓ , $\mathbf{v} = 2\mathbf{i} - 5\mathbf{j} - 3\mathbf{k}$ is parallel to $\mathcal{P}$, and $\mathcal{P}$ also contains $\mathbf{B}(1, 1, 0)$.

Since $\mathbf{A}(-1, 2, 1)$ and $\mathbf{B}(1, 1, 0)$ lie on $\mathcal{P}$, the vector

$$\mathbf{w} = \overrightarrow{\mathbf{AB}} = (1 - (-1))\mathbf{i} + (1 - 2)\mathbf{j} + (0 - 1)\mathbf{k} = 2\mathbf{i} - \mathbf{j} - \mathbf{k}$$

is parallel to the plane $\mathcal{P}$.

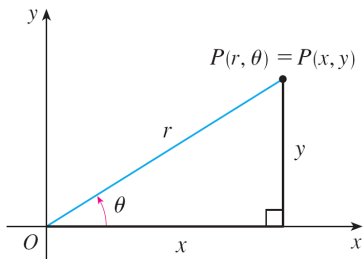
Therefore, a normal vector of the plane $\mathcal{P}$ can be taken as

$$\vec{N} = \mathbf{v} \times \mathbf{w} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -5 & -3 \\ 2 & -1 & -1 \end{vmatrix} = 2\mathbf{i} - 4\mathbf{j} + 8\mathbf{k}.$$

Together with the fact that $\mathbf{B}(1, 1, 0)$ lies on $\mathcal{P}$, an equation of the plane $\mathcal{P}$ is

$$\begin{aligned} \langle \mathbf{x} - 1, \mathbf{y} - 1, \mathbf{z} - 0 \rangle \cdot \langle 2, -4, 8 \rangle &= 0 \\ \iff 2x - 2 - 4y + 4 + 8z &= 0 \\ \iff x - 2y + 4z + 1 &= 0. \end{aligned}$$

Polar coordinates.



Instead of using Cartesian coordinates (x, y) , any point in $\mathbb{R}^2$ can be represented by **polar coordinates** (r, θ) , where

- r is the distance between the origin and the point,
- θ is the angle from the positive x -axis to the straight line joining the origin and the point.

Equations relating (x, y) and (r, θ) :

$$x = r \cos \theta, \quad y = r \sin \theta,$$

$$r^2 = x^2 + y^2, \quad \tan \theta = \frac{y}{x}.$$

Q05.6. (a) Consider the point $P(2, -2\sqrt{3})$ in Cartesian coordinates. Rewrite the point P in polar coordinates. If possible, give an alternative answer using another ordered pair of numbers.

(b) Consider the point $Q(6, \frac{3\pi}{4})$ in polar coordinates. Rewrite the point Q in Cartesian coordinates.

Q05.6. (a) Consider the point $P(2, -2\sqrt{3})$ in Cartesian coordinates. Rewrite the point P in polar coordinates. If possible, give an alternative answer using another ordered pair of numbers.

Solution:

$$r = \sqrt{x^2 + y^2} = \sqrt{2^2 + (-2\sqrt{3})^2} = \sqrt{4 + 12} = 4.$$

$$\tan \theta = \frac{y}{x} = \frac{-2\sqrt{3}}{2} = -\sqrt{3} \implies \theta = \frac{2\pi}{3} \text{ or } \frac{5\pi}{3}.$$

But $\theta = \frac{2\pi}{3}$ is rejected, since $P(2, -2\sqrt{3})$ lies in the fourth quadrant.

Thus in polar coordinates, P is given by $(r, \theta) = (4, \frac{5\pi}{3})$.

Another polar representation of P is: $(r, \theta) = (4, -\frac{\pi}{3})$.

(b) Consider the point $\mathbf{Q}(6, \frac{3\pi}{4})$ in polar coordinates.

Rewrite the point $\mathbf{Q}$ in Cartesian coordinates.

Solution:

$$x = r \cos \theta = 6 \cos \frac{3\pi}{4} = 6 \cdot \frac{-1}{\sqrt{2}} = -3\sqrt{2}.$$

$$y = r \sin \theta = 6 \sin \frac{3\pi}{4} = 6 \cdot \frac{1}{\sqrt{2}} = 3\sqrt{2}.$$

Thus in Cartesian coordinates, $\mathbf{Q}$ is given by

$$(\mathbf{x}, \mathbf{y}) = (-3\sqrt{2}, 3\sqrt{2}).$$

Q05.7. For the following equations in Cartesian coordinates, rewrite them in terms of polar coordinates:

(i) $\frac{x^2}{9} + \frac{y^2}{4} = 1.$

(ii) $y^2 = 6x.$

Solution: (i)

$$\begin{aligned}\frac{x^2}{9} + \frac{y^2}{4} = 1 &\iff \frac{(r \cos \theta)^2}{9} + \frac{(r \sin \theta)^2}{4} = 1 \\ &\iff 4r^2 \cos^2 \theta + 9r^2 \sin^2 \theta = 36.\end{aligned}$$

(ii)

$$y^2 = 6x \iff (r \sin \theta)^2 = 6r \cos \theta$$

$$\iff r^2 \sin^2 \theta = 6r \cos \theta$$

$$\iff r(r \sin^2 \theta - 6 \cos \theta) = 0$$

$$\iff r = 0 \text{ or } r \sin^2 \theta - 6 \cos \theta = 0$$

$$\iff r \sin^2 \theta - 6 \cos \theta = 0,$$

since in the second last line, the solution of the first equation $r = 0$ is included in those of the second equation with $(r, \theta) = (0, \frac{\pi}{2})$.

Q05.8. A point $\mathbf{P}$ in $\mathbb{R}^2$ is such that both the Cartesian coordinates and polar coordinates of $\mathbf{P}$ are given by the same ordered pair of numbers. Determine the locus of $\mathbf{P}$. First give your answer in terms of polar coordinates, and then give also your answer in terms of Cartesian coordinates.

Solution: Let the Cartesian coordinates of $\mathbf{P}$ be (x, y) , and the polar coordinates of $\mathbf{P}$ be (r, θ) . Then we have

$$(x, y) = (r, \theta) \iff \begin{cases} x = r, & (1) \\ y = \theta. & (2) \end{cases}$$

From (1), we have

$$x = r \implies x^2 = r^2 = x^2 + y^2 \implies y^2 = 0 \implies y = 0.$$

Substituting $y = 0$ into (2), we get

$$\theta = 0,$$

which corresponds to all the points on the positive x -axis (including the origin).

Now observe that all these points satisfy (1) and (2).

Thus the locus of P is given by polar equation $\theta = 0$.

Then the locus of **P** in Cartesian coordinates is given by

$$y = 0 \text{ and } x \geq 0.$$

Q05.9. (a) For each part below, sketch the curve defined by the given polar equation (i.e. equation in polar coordinates):

(i) $r = 2(1 - \cos \theta)$;

(ii) $r = 6 \sin \theta$.

(b) For both parts (i) and (ii), rewrite the equation in Cartesian coordinates.

(a) (i) $r = 2(1 - \cos \theta)$

Solution:

(a) (ii) $r = 6 \sin \theta$.

Solution:

(b) For both parts (i) and (ii), rewrite the equation in Cartesian coordinates.

(i) $r = 2(1 - \cos \theta)$

(ii) $r = 6 \sin \theta$.

Q05.10 to Q05.14 are on Section 12.1 -12.3.

Vector-valued functions of one variable

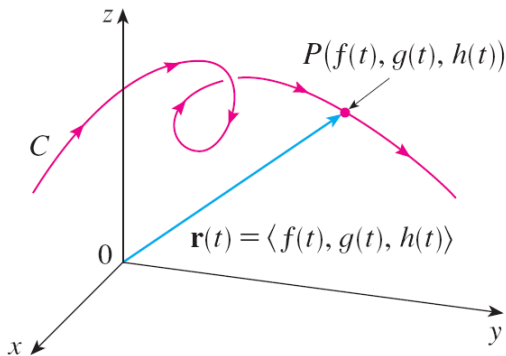
A vector-valued function of one variable is of the form

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$$

or

$$\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$$

for some scalar function ***f***, ***g*** and ***h*** (called the component functions of ***r***).



As $\mathbf{t}$ varies, $\mathbf{r}(\mathbf{t})$ traces out a curve $\mathbf{C}$ (in $\mathbb{R}^2$ or $\mathbb{R}^3$), and we say that $\mathbf{r}(\mathbf{t})$ is a parametrization of $\mathbf{C}$.

Q05.10. Consider the vector function

$$\mathbf{v}(t) = (3t^2 + 2t)\mathbf{i} + \pi \sin(\pi t)\mathbf{j} + e^t\mathbf{k}.$$

(i) Evaluate the derivative $\mathbf{v}'(t)$. Find also $\mathbf{v}'(2)$.

(ii) Evaluate the indefinite integral $\int \mathbf{v}(t) dt$.

(iii) Evaluate the definite integral $\int_0^1 \mathbf{v}(t) dt$.

Consider the vector function

$$\mathbf{v}(t) = (3t^2 + 2t)\mathbf{i} + \pi \sin(\pi t)\mathbf{j} + e^t\mathbf{k}.$$

(i) Evaluate the derivative $\mathbf{v}'(t)$. Find also $\mathbf{v}'(2)$.

Solution:

$$\mathbf{v}(t) = (3t^2 + 2t)\mathbf{i} + \pi \sin(\pi t)\mathbf{j} + e^t\mathbf{k}.$$

Q05.11. Consider the curve in $\mathbb{R}^3$ given by $\mathbf{r}(t) = (\ln t)\mathbf{i} + \frac{t-1}{t+3}\mathbf{j} + (t \ln t)\mathbf{k}$. Find parametric equations of the tangent line to the curve at the point where $t = 1$.

Solution:

Q05.12. A particle moves in $\mathbb{R}^3$ such that its velocity at time t is given by

$$\mathbf{v}(t) = (6t^2 + 4t)\mathbf{i} + (8t^3 + 2t)\mathbf{j} + \frac{1}{1+t^2}\mathbf{k}.$$

If the position of the particle at time $t = 0$ is $\mathbf{i} + 2\mathbf{j} + \pi\mathbf{k}$. Find the position of the particle at time $t = 1$.

[Remark: The position $\mathbf{r}(t)$ and velocity $\mathbf{v}(t)$ (as vector functions of time t) of the particle satisfy $\mathbf{r}'(t) = \mathbf{v}(t)$.]

Q05.13. Two particles move in $\mathbb{R}^3$ such that their positions at time t are given by:

$$\mathbf{r}_1(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k},$$

$$\mathbf{r}_2(t) = (1 + 2t)\mathbf{i} + (1 + 6t)\mathbf{j} + (1 + 14t)\mathbf{k}.$$

- (a) Do the particles collide? Justify your answer.
- (b) Do their paths intersect? Justify your answer.

Solution: (a) For particles to collide, we equate the two vector functions using the *same* parameter t :

$$\mathbf{r}_1(t) = \mathbf{r}_2(t).$$

Equating the three components, we get

$$t = 1 + 2t, \quad (1)$$

$$t^2 = 1 + 6t, \quad (2)$$

$$t^3 = 1 + 14t. \quad (3)$$

This system does not have solution. To see this, from (1), we get $t = -1$. But $t = -1$ does not satisfy (2), since $(-1)^2 = 1 \neq 1 + 6(-1) = -5$.

(Similarly $t = -1$ does not satisfy (3).)

So, we conclude that the 2 particles do not collide.

(b) For the paths to intersect, we equate the two vector functions using *different* parameters $\mathbf{s}$ and $\mathbf{t}$:

$$\mathbf{r}_1(\mathbf{t}) = \mathbf{r}_2(\mathbf{s}).$$

Equating the components, we get

$$\mathbf{t} = \mathbf{1} + \mathbf{2s}, \quad (1)$$

$$\mathbf{t}^2 = \mathbf{1} + \mathbf{6s}, \quad (2)$$

$$\mathbf{t}^3 = \mathbf{1} + \mathbf{14s}. \quad (3)$$

This system has two solutions given by

$$\mathbf{s} = \mathbf{0}, \mathbf{t} = \mathbf{1} \quad \text{and} \quad \mathbf{s} = \frac{\mathbf{1}}{\mathbf{2}}, \mathbf{t} = \mathbf{2}.$$

To see this, we substitute (1) into (2), and we get

$$(1 + 2s)^2 = 1 + 6s \implies 4s^2 - 2s = 0$$

$$\implies 2s(2s - 1) = 0 \implies s = 0, \frac{1}{2}.$$

Substituting $s = 0$ into (1), we get $t = 1$.

Substituting $s = \frac{1}{2}$ into (1), we get $t = 1 + 2 \cdot \frac{1}{2} = 2$.

Thus we get two potential solutions given by

$$s = 0, t = 1 \quad \text{and} \quad s = \frac{1}{2}, t = 2.$$

Now we substitute $\mathbf{s} = \mathbf{0}$, $\mathbf{t} = \mathbf{1}$ into (1), (2), (3) to see that $\mathbf{s} = \mathbf{0}$, $\mathbf{t} = \mathbf{1}$ is indeed a solution of the system of the three equations. (This amounts to checking that

$$\mathbf{1} = \mathbf{1} + \mathbf{2} \cdot \mathbf{0}, \quad \mathbf{1}^2 = \mathbf{1} + \mathbf{6} \cdot \mathbf{0}, \quad \mathbf{1}^3 = \mathbf{1} + \mathbf{14} \cdot \mathbf{0}.)$$

Similarly, we substitute $\mathbf{s} = \frac{\mathbf{1}}{\mathbf{2}}$, $\mathbf{t} = \mathbf{2}$ into (1), (2), (3) to see that $\mathbf{s} = \frac{\mathbf{1}}{\mathbf{2}}$, $\mathbf{t} = \mathbf{2}$ is indeed a solution of the system of the three equations. (This amounts to checking that

$$\mathbf{2} = \mathbf{1} + \mathbf{2} \cdot \frac{\mathbf{1}}{\mathbf{2}}, \quad \mathbf{2}^2 = \mathbf{1} + \mathbf{6} \cdot \frac{\mathbf{1}}{\mathbf{2}}, \quad \mathbf{2}^3 = \mathbf{1} + \mathbf{14} \cdot \frac{\mathbf{1}}{\mathbf{2}}.)$$

Q05.14. Find the arc length of the curve

$$x = 2t - 2 \sin t, \quad y = 2 - \sqrt{2} \cos t, \quad z = \sqrt{2} \cos t$$

for $0 \leq t \leq 4\pi$.

Solution: From the equation of given curve, we have

$$x = a(t - \sin t) \implies \frac{dx}{dt} = a(1 - \cos t),$$
$$y = a(1 - \cos t) \implies \frac{dy}{dt} = a \sin t.$$

Thus the arc length of the curve is given by

$$\begin{aligned} L &= \int_0^{4\pi} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \\ &= \int_0^{4\pi} \sqrt{a^2(1 - \cos t)^2 + a^2 \sin^2 t} dt \\ &= a \int_0^{4\pi} \sqrt{1 - 2 \cos t + \cos^2 t + \sin^2 t} dt \\ &= a \int_0^{4\pi} \sqrt{2 - 2 \cos t} dt = 2a \int_0^{4\pi} \sqrt{\sin^2\left(\frac{t}{2}\right)} dt. \end{aligned}$$

Note that for $0 \leq t \leq 2\pi$, we have $\sin(\frac{t}{2}) \geq 0$; and for $2\pi \leq t \leq 4\pi$, we have $\sin(\frac{t}{2}) \leq 0$. Therefore,

$$\begin{aligned}\int_0^{4\pi} \sqrt{\sin^2(\frac{t}{2})} dt &= \int_0^{4\pi} \left| \sin(\frac{t}{2}) \right| dt \\&= \int_0^{2\pi} \sin(\frac{t}{2}) dt + \int_{2\pi}^{4\pi} -\sin(\frac{t}{2}) dt \\&= \left[-2 \cos(\frac{t}{2}) \right]_0^{2\pi} + \left[2 \cos(\frac{t}{2}) \right]_{2\pi}^{4\pi} \\&= (-2) \cdot (-1) - (-2) \cdot 1 + 2 \cdot 1 - 2 \cdot (-1) \\&= 8.\end{aligned}$$

Hence the arc length of the curve is $2a \cdot 8 = 16a$.